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Source: *Journal of Consumer Research*, Jun., 1993, Vol. 20, No. 1 (Jun., 1993), pp. 157-169

Published by: Oxford University Press

Stable URL: <https://www.jstor.org/stable/2489208>

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Age Differences in Consumers' Search for Information: Public Policy Implications

CATHERINE A. COLE
SIVA K. BALASUBRAMANIAN*

We investigated whether consumers in their sixties (or older) can use nutritional information as accurately as younger consumers in a pair of studies, the first conducted in a supermarket setting, the second in a laboratory. Both studies indicate that, when shoppers are instructed to select a cereal according to specific nutritional criteria, elderly subjects are less likely than younger subjects to search intensely and to select an appropriate cereal. In the laboratory setting, however, the age-related differences diminished when subjects wrote down all the nutritional information acquired during their search. Age-related changes in information-processing ability may explain the findings. Implications for public policy are discussed.

In 1990, when Congress updated the 1973 Food and Drug Administration labeling rule, it required food manufacturers to put more nutritional information on their labels. Although this rule was updated to make it easier for consumers to choose more nutritious foods, the literature on age differences in problem solving suggests that older consumers may not successfully search for and use this information. To investigate whether consumers in their sixties (or older) can use nutritional information as accurately as younger consumers, we studied how different age groups utilize the information printed on the sides of cereal boxes. We also observed how supermarket shoppers and subjects in a computer laboratory solved a nutritional information problem: find a cereal that has less than 200 milligrams of sodium and at least 2 grams of fiber. We suspected that age differences would emerge for two reasons. First, the consumer behavior literature reports that as people age they tend to limit the amount of information they obtain about products prior to a brand choice decision (Beatty and Smith 1987). Second, the developmental literature suggests that an age-associated decline in working-memory capacity may adversely affect the ability of some older adults to solve problems.

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AGE DIFFERENCES IN PROBLEM SOLVING

When we solve a problem, we must rely on working memory, a critical component of an individual's information-processing system consisting of a limited-capacity work space that can be divided between storage and processing functions. Together, these memory functions play a crucial role in cognitive tasks such as learning and reasoning (Baddeley 1990). Because of its limited capacity, this system must trade off holding information in short-term storage with manipulating information to solve the problem.

Although a limited working memory constrains the amount of information any person can process while solving a problem, elderly adults face an especially limited working memory. Wright (1981), for example, varied the number of digits subjects had to keep in memory in order to perform simple addition problems and found that older subjects had increasing difficulty as the digit load grew.

Salthouse and Prill (1987) found sizable age differences in performance on series-completion problems (such as "what number follows: 2, 4, 8, . . . ?"). Because age did not appear to affect accuracy with simple arithmetic skills, they attributed observed differences to age-associated difficulties in progressive abstraction and flexibility. To solve series-completion problems people must understand how elements in the series are interrelated. Older adults appear to build less stable relational structures than do younger adults. Limited working-memory capacity may lead to unreliable structures because the products of earlier processing can be lost.

This age-related decline in working-memory capacity seems to constrain performance especially on unfamiliar

iar tasks (Salthouse 1991). However, when elderly adults solve familiar problems, they often perform as well as or better than younger adults (Charness 1981). Although working-memory capacity diminishes with age, older adults may learn to compensate for such loss when they are familiar with the problem task (Smith and Moschis 1990).

Age Differences in Consumer Search Problems

Some empirical evidence suggests that the information search behavior of older consumers is different from the search behavior of younger consumers. Factors like experience with the product category, familiarity with the brands, and information-processing ability may explain this difference. A survey of automobile buyers found that older consumers searched less and were more satisfied with their previous choices than other clusters of customers (Furse, Punj, and Stewart 1984). Age differences may have emerged because older consumers had more experience to rely on than younger consumers as they selected a car.

In another study, when selecting unfamiliar brands from four different product categories, older housewives considered fewer attributes and alternatives than younger housewives (Schaninger and Sciglimpaglia 1981). Older consumers with diminished working-memory capacity cannot hold and manipulate numerous alternatives in memory. As a result, they may not search for as much information as younger consumers. They may also make inferior choices because they can neither form stable brand evaluations nor process new information to reevaluate their impression of different brands. Such consumers are especially likely to employ a "satisficing" strategy; that is, they may engage in relatively superficial searches for solutions, inspecting only a few requirements and considering alternatives only until an acceptable solution is found (Janis and Mann 1977). Such consumers may feel less satisfied with their choice than others who thoroughly process the available information on all alternatives. Although no literature explicitly investigates whether consumer characteristics (such as age) affect the tendency to satisfice, it is reasonable to infer that, when older consumers solve new problems, they may limit their search, given their information-processing limitations.

STUDY 1

Overview

The purpose of the field study was to determine if age differences in search intensity and search outcome emerge when consumers solve either familiar or new search problems in the supermarket aisle. In this study, we interviewed a total of 79 consumers in three chain grocery stores in a midwestern university town.

Shoppers were randomly assigned to one of two tasks. In the *unconstrained* task condition, we simply observed shoppers' search behavior and brand choice in the cereal aisle and later interviewed them. This task should be familiar to shoppers, given that we did not intervene to constrain or structure their search problem. In the *constrained* task condition, however, we defined the shoppers' search problem by giving them a specific nutrition-based decision rule for choosing a cereal. We also observed these shoppers and later asked them questions. As compensation for participating, every shopper was offered a \$1.00 coupon for the brand of cereal they had chosen.

Hypotheses

Under normal grocery-shopping conditions few age differences should emerge. Grocery shopping is a familiar task, and even if elderly adults experience cognitive changes, they may have learned to compensate for them in solving the problems they frequently encounter in the supermarket. Consistent with an earlier study, we expect to find that shoppers consider only part of the available information and spend a minimal amount of time selecting a brand (Dickson and Sawyer 1990). We expect age differences to emerge, however, when we ask shoppers to consider new criteria (i.e., the decision rule) in making their selection. The reduced information-processing ability of elderly shoppers may make them more likely to act as satisficers, less likely to search intensely, and more likely to choose incorrectly than younger adults.

H1: There will be an age-by-task interaction such that (a) there are no age differences in search intensity, search outcome, and satisficing behavior in the unconstrained task, but (b) there are age differences in search intensity, search outcome, and satisficing in the constrained task condition.

Previous studies (e.g., Punj and Staelin 1983) indicate that, as the amount of search prior to a decision increases, consumers' satisfaction with their purchase increases. Our research extends this previous work by studying observed (not self-reported) measures of search for a frequently purchased grocery item (not a durable good).

H2: As the search becomes more intense, consumer satisfaction with the final choice will increase.

Independent Variables

Task. Shoppers who entered the cereal aisle were randomly assigned to either an unconstrained or constrained task. In the former, the interviewer approached each shopper after s/he had chosen a cereal and solicited

the shopper's participation in a university research project.¹ In the latter, the interviewer approached each shopper who entered the cereal aisle, explained that she was asking shoppers to select a cereal that met certain nutritional criteria, and asked whether the shopper would like to participate. Most shoppers (81 percent) agreed to participate. Next the interviewer explained the decision rule: shoppers could select any cereal that had less than 200 milligrams of sodium and at least 2 grams of fiber in a serving. The interviewer handed the shopper a piece of paper with this rule typed on it. Before the subject began searching for a cereal that met the decision criteria, the interviewer showed the shopper a nutritional panel cut out from a cereal box (with all identifying features blacked out) and asked the shopper (1) to read aloud the sodium and fiber content listed on the panel and (2) to indicate whether the cereal met the criteria. This procedure helped us determine whether participants could read the nutritional information on the panel and whether they understood the decision rule.² The shopper then proceeded to select a box of cereal and was reinterviewed by the person who had originally intercepted the shopper.

Age. We divided subjects into two groups. The ages of the younger participants ranged between 20 and 59 years (average age was 36); those of the older subjects ranged from 60 to 89 years (average age was 68).

Dependent Variables

The operational definitions of the variables for search intensity and outcome are listed in Table 1. We assigned an observer (equipped with a clipboard to appear to be taking inventory) to the cereal aisle to unobtrusively collect data on several variables for each subject.³ The observer recorded the brand names of cereals that the shopper looked at (Boxes). If the shopper picked up a box, the observer placed a check mark under the recorded brand name (Npick); if the shopper turned the box to inspect the nutritional panel, the brand name was circled (Panels); when the shopper placed a package in the cart, the observer crossed out the brand name chosen (Choice).

Results

Preliminary Data Analysis. We compared the elderly and younger adults on several demographic and

TABLE 1
DEPENDENT VARIABLES: OPERATIONAL DEFINITIONS

Dimension and variable name	Operational definition
Study 1:	
Search intensity:	
Boxes	Number of cereal boxes inspected.
Npick	Number of cereal boxes picked up off the shelf.
Panels	Percentage of boxes for which nutritional panels were inspected.
Search outcome:	
Satisfac	Two-item Likert scale to assess degree of satisfaction with the cereal brand chosen ($r = .63$).
Choice	Value = 1 if the brand chosen met the decision rule (constrained condition only); value = 0 otherwise.
Study 2:	
Search intensity:	
Ninfo	Total number of pieces of information that the subject asked for while conducting the search.
Ndatts	The number of different attributes the subject considered.
Nbrands	Number of brands the subject inspected.
Search outcome:	
Satisfac	Satisfaction measure based on six-item Likert scale ($\alpha = .75$).
Choice	Value = 1 if the brand chosen met the decision rule; value = 0 otherwise.

attitudinal variables. Based on our single-item measure, younger adults had completed more schooling than elderly adults ($t = 2.07$, $df = 75$, $p < .05$).⁴ There were no significant age differences for measured variables such as cereal knowledge (three items; coefficient alpha = .71) and usage (three items; coefficient alpha = .73). However, elderly shoppers expressed more loyalty for the brands they selected than younger adults ($t = 2.08$, $df = 75$, $p < .05$) on our brand-loyalty scale (two items; interitem correlation = .69). Because earlier research demonstrates that brand loyalty negatively affects consumer search (Punj and Staelin 1983), we used brand loyalty as a covariate in all our subsequent analyses. There were no significant task or age effects on our single-item measure of motivation to find the best cereal, nor did these factors affect the amount of time shoppers spent selecting a cereal. Thus, it appears that both elderly and younger adults devoted similar effort in cereal selection.

¹We pretested the task manipulation four times in the supermarket. Shoppers in the unconstrained condition who passed down the aisle without selecting a cereal were not included in the study. Shoppers who bought more than one package of cereal were asked to answer questions about the first brand selected; in this case, search-intensity measures were based on the shopper's actions before making the first choice.

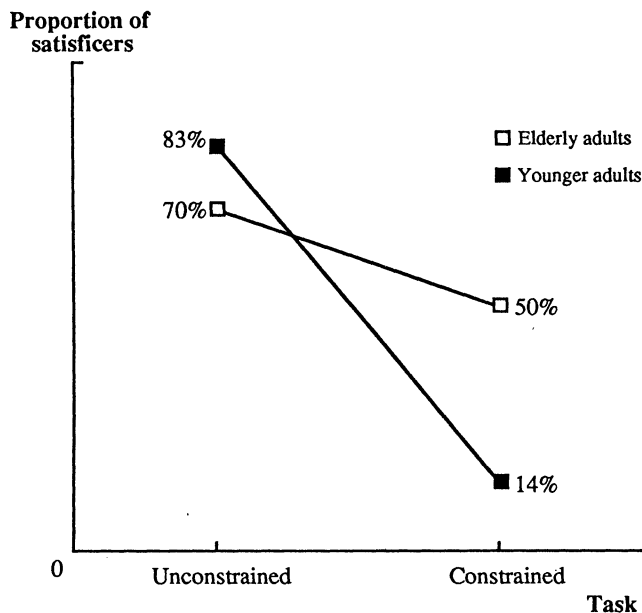
²We eliminated from our analysis two subjects (one older, one younger) who could not indicate whether the cereal met the criteria after three trials.

³In a pretest with two observers, the interobserver agreement for recorded data on observed variables was found to be 93 percent.

⁴We did not use education as a covariate because of difficulties associated with comparing individuals of different ages; e.g., the high school education of a 25-year-old person is intrinsically very different from the high school education of another who is 75 years old.

FIGURE 1

STUDY 1: SATISFICERS AS A FUNCTION OF AGE AND TASK

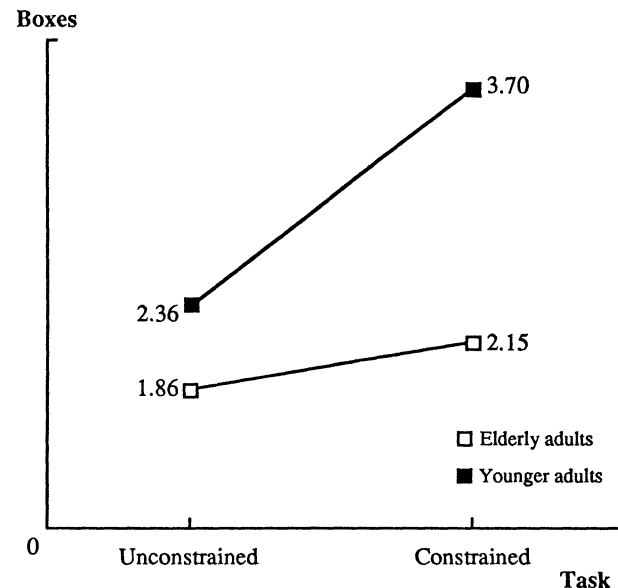


Satisficing Behavior. We defined a satisficer as a person who selected the first cereal box inspected. The proportions of satisficers are reported in Table 2. Using logistic regression with brand loyalty as a covariate, we analyzed whether age, task, or the interaction between the two affected satisficing behavior. The interaction, which was significant ($\chi^2 = 5.55$, $df = 1$, $p < .05$), is diagrammed in Figure 1. Younger adults were significantly less likely to be satisficers in the constrained than in the unconstrained condition ($\chi^2 = 14.42$, $df = 1$, $p < .01$), but the task did not significantly affect whether elderly adults were satisficers or not. Another way of looking at the interaction is to note that, in the unconstrained condition, elderly and younger adults were equally likely to choose without an extensive search, but in the constrained condition elderly shoppers were more likely to limit their search than younger shoppers ($\chi^2 = 4.41$, $df = 1$, $p < .03$).

Search Intensity. For this analysis, the cell means are reported in Table 2; the analysis of covariance (ANCOVA) results are reported in Table 3. In the overall multivariate analysis of covariance (MANCOVA) for search intensity, we used brand loyalty as a covariate and the following as dependent variables: the number of boxes inspected, the number of boxes picked up, and the percentage of panels that the subject inspected for nutritional information. The analysis shows a significant age effect (Wilks's lambda = .84, $F(3,70) = 4.42$, $p < .01$), a significant task effect (Wilks's lambda = .28, $F(3,70) = 58.63$, $p < .01$), and a significant age-by-task interaction (Wilks's lambda = .87, $F(3,70) = 3.51$, p

FIGURE 2

STUDY 1: NUMBER OF BOXES INSPECTED AS A FUNCTION OF AGE AND TASK



$< .02$). We then inspected the univariate ANCOVAs and found all the interactions to be significant (see Figs. 2–4).

For the number of boxes inspected, there were no age differences in the unconstrained condition, but in the constrained condition, younger adults inspected significantly more boxes than elderly adults ($p < .05$). For the number of boxes that the shopper actually picked up, the same pattern emerged: no significant age differences in the unconstrained condition, but significant age differences in the constrained condition ($p < .01$). These interactions occur because, as predicted in Hypothesis 1, age does not affect how shoppers perform familiar tasks, but it does affect how shoppers perform novel tasks. Thus, younger adults search more than elderly adults in the constrained condition. The results from the analysis of the age-by-task interaction for the percentage of nutritional panels inspected are somewhat different: there are no significant age differences in the unconstrained condition, but in the constrained condition there are significant age differences even though both age groups inspected a significantly greater percentage of panels.

Search Outcome. For this analysis, the cell means and proportions are reported in Table 2. We first assessed how satisfied subjects were with their choice using ANCOVA (with brand loyalty as a covariate) to test the effects of age and task (Table 3). Younger adults were significantly more satisfied with their brand choice than elderly adults. We later analyzed whether subjects correctly applied the decision rule in the constrained task. A significantly higher proportion of the younger shop-

FIGURE 3

STUDY 1: NUMBER OF BOXES PICKED UP AS A FUNCTION OF AGE AND TASK

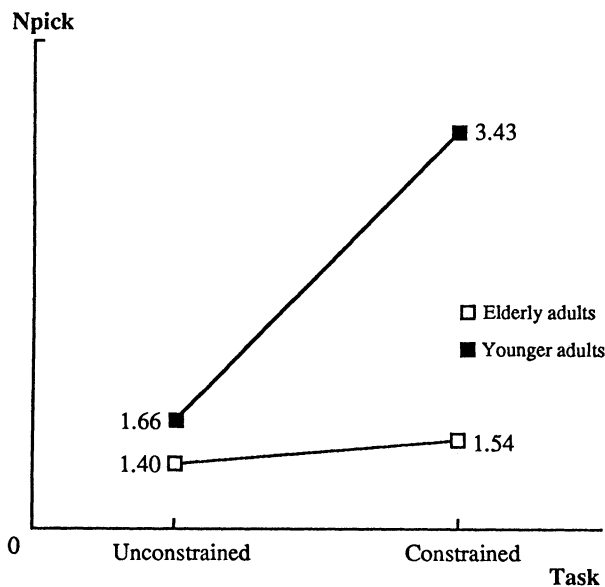
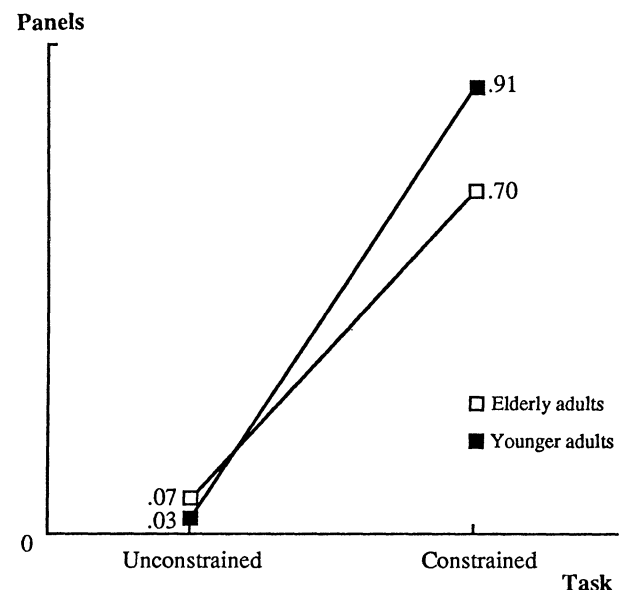


FIGURE 4

STUDY 1: PROPORTION OF NUTRITIONAL PANELS INSPECTED AS A FUNCTION OF AGE AND TASK



pers (91 percent) than of the elderly shoppers (58 percent) were successful in this regard ($\chi^2 = 3.86$, $df = 1$, $p < .05$).

Search Intensity and Search Outcome. Finally, to test the hypothesized positive relationship between search intensity and search outcome, we examined the correlations between the search-intensity measures and satisfaction. We found that satisfaction increased as the number of boxes inspected increased ($r = .24$, $p < .04$). Although we have some support for Hypothesis 2, it appears that the magnitude of the relationship between search intensity and satisfaction is at best moderately strong for routine purchases.

Summary

In normal conditions, consumers do not search much for information about the cereals they buy. Because most supermarket shoppers compare only two or three brands before making a choice, no age differences emerge. However, when given a novel decision task involving nutritional information, younger adults search more intensely and are more likely to select an acceptable cereal than elderly adults. These results suggest that elderly shoppers will be at a disadvantage when they make decisions that require them to use nutritional information in a new way.

Although this study supports our hypothesis that age differences in search behavior and information usage are likely to emerge when consumers solve new search problems, it does not tell us why these differences occur. Even though the elderly shoppers said they were mo-

tivated to process the necessary nutritional information, seemed to expend as much effort as younger adults, and appeared to understand the decision rule, they neither searched as intensely nor solved the problem as well as younger adults. The need to explore rationales for these findings motivates additional research. Further, note that the supermarket setting of this study made brand name information available to shoppers. Therefore, we did not know whether elderly shoppers searched less because they were already familiar with the sodium and fiber content of various brands or because they had trouble processing new information. To avoid this problem, the brand names in the next study were disguised.

STUDY 2

Overview

The purpose of our second laboratory study was to determine whether the age differences in search behavior and search outcome stem from age-associated changes in information processing. In addition, we explored whether the differences could be diminished by instructing consumers to write down the information they acquire during their search.

Hypotheses

Our study compares unaided and aided search tasks. In the unaided task, the decision makers search for information and store it in memory as it is acquired. If

TABLE 2
MEANS AND PROPORTIONS

Dimension/dependent variables	Cell means and proportions			
	Younger adults		Elderly adults	
	U ^a	C ^a	U ^a	C ^a
Study 1:				
Satisficing behavior:				
Satisficer proportion (%)	83.0	14.0	70.0	50.0
Search intensity:				
Boxes ^b	2.36	3.70	1.86	2.15
Npick ^b	1.66	3.43	1.40	1.54
Panels ^b	.03	.91	.07	.70
Search outcome:				
Satisfac ^b	9.51	9.20	8.78	8.79
Choice proportion (%) ^c	NA	91.0	NA	58.0
Study 2:				
Satisficing behavior:				
Satisficer proportion (%)	5.3	.0	9.5	26.3
Search intensity:				
Ninfo	18.32	20.90	9.80	5.00
Ndatts	5.37	5.43	4.25	2.63
Nbrands	4.84	4.76	3.95	2.74
Search outcome:				
Satisfac	23.11	21.67	19.90	20.05
Choice proportion (%)	89.5	95.2	81.0	73.7

^aU is the unconstrained task; C is the constrained task.

^bThese means are adjusted for the brand-loyalty covariate.

^cThis variable was calculated only for the constrained task in which shoppers were given a decision rule; NA = not applicable.

^dA is the aided task; Ua is the unaided task.

all subjects face unfamiliar brands, age might affect measures of the search process because elderly consumers, given their constrained working-memory capacity, cannot obtain, store, and process new information as effectively as younger adults. During the aided task, decision makers search for information and write it down so that it is externally available when they make a decision. By writing information down, decision makers eliminate the need to store and process newly acquired information simultaneously.

Because the aid reduces the strain of storage on the elderly adult's working memory, it should free more capacity for processing information. Thus, elderly adults should be less likely to act as satisficers, search more intensely, and arrive at a better search outcome in the aided than in the unaided condition. We anticipate, however, that unaided search should not tax younger adults' working-memory capacity; the aid should not affect how much younger adults search or

the quality of the search outcome. Thus, age differences in satisficing, search intensity, and search outcome will diminish in aided search.

H3: There will be an age-by-task interaction for the satisficing, search-intensity, and outcome quality variables. In the unaided search task, elderly adults will be more likely to satisfice, will search less intensely, and will have less successful search outcomes than younger adults. In the aided task, there will be no age differences in satisficing, search intensity, and outcome quality.

H4: Age and working-memory capacity will correlate negatively.

We also tested Hypothesis 2 from Study 1, which predicts that as the search becomes more intense consumer satisfaction with the final choice will increase.

We used data on five actual brands of cereal in the experiment. Each brand was characterized by information about nine attributes: ingredients, description, taste, *Consumer Reports'* nutritional rating, calories, and sodium, fiber, sugar, and protein content. We collected the data about sodium, fiber, nutritional rating, taste, and sugar from *Consumer Reports* (1989). Data for the remaining attributes were obtained from the actual packages of the five brands. Throughout the experiment the brands were merely labeled with the letters A, B, C, D, and E. To keep subjects from trying to guess what the cereals actually were, we gave a cover story that these were new brands that would soon be introduced into the local market.

Description of the Computerized Part of the Study

We used an adapted version of Search Monitor (Brucks 1988), an interactive program that does not require previous computer experience; participants simply select alternatives from menus listed on the computer screen.⁵

The first computer screen contained instructions about the keyboard. All subjects were then presented with a series of simple arithmetic problems (e.g., how much is 34 – 33?), for which they were to type in the correct number. This typing test served to exclude from the study anyone who could not see or use the keyboard.

When a participant responded correctly to three arithmetic problems in a row, the program proceeded to a practice search exercise in which subjects could seek information about three fast-food restaurants and compare them across three attributes. This computer exercise familiarized subjects with how to conduct in-

⁵A growing body of gerontology and psychology research demonstrates that computers are an effective way to test or train elderly adults for several performance tasks (e.g., Stones and Kozma 1989).

TABLE 3
ANCOVA AND ANOVA RESULTS

		Dependent variable							
Source of variation	df	Boxes ^a		Npick ^a		Panels ^a		Satisfac ^a	
		SS	F	SS	F	SS	F	SS	F
Study 1:									
Age (A)	1	18.30	6.15*	20.12	12.65*	.13	2.23	5.51	5.52*
Task (T)	1	12.34	4.15*	16.75	10.53*	10.49	177.60*	.39	.39
A × T	1	17.71	5.96*	11.81	7.43	.25	4.28*	.44	.44
Corrected total	76								
		Dependent variable							
		Ninfo		Ndatts		Nbrands		Satisfac	
		SS	F	SS	F	SS	F	SS	F
Study 2:									
Age (A)	1	2,939.00	34.70*	75.60	21.17*	41.94	31.16*	115.60	5.40*
Task (T)	1	24.00	.28	11.96	3.39**	8.25	6.12*	8.31	.39
A × T	1	269.00	3.17**	13.88	3.89*	6.33	4.70*	12.55	.59
Corrected total	79								

^aBrand loyalty is a covariate. SS = sum of squares; F = F-statistic.

* $p < .05$.

** $p < .10$.

formation search in order to decide on a choice alternative. Once participants selected a restaurant, the computer screen directed them to contact the researcher, who encouraged questions and provided a printed handout. The handout explained that (1) the purpose of the main search task was to select one of five brands of cereal to take home, (2) the cereal selected should satisfy a dietician's decision rule (the same rule as used in study 1), (3) participants could request as little or as much information as they wanted, and (4) at the end of the study participants would receive a gift pack of the brand they had selected in their search.

Before beginning the cereal-shopping task, subjects completed another short computer exercise to verify their understanding of the dietician's decision rule. This exercise presented subjects with a hypothetical brand of cereal (e.g., Brand Z) and the sodium and fiber level in the cereal. Subjects had to judge whether the cereal met the dietician's criteria. If they were wrong, the next screen explained why they were wrong and gave them a new choice. The purpose of the exercise was to bring all subjects, regardless of age, to an equal level of familiarity with the decision rule. Once participants could apply the decision rule correctly three consecutive times, additional computer screens reinforced the information in the handout. Then the participants began shopping.

Once a subject obtained a piece of information about a brand, the menu on the screen displayed four alter-

natives: (1) to quit the shopping trip and order a cereal, (2) to search for information on the same attribute for a different brand, (3) to search for information on a different attribute for the same brand, or (4) to search for information on a different attribute for a different brand. Subjects continued to search until they selected the option of ending the shopping trip and ordering a cereal.

Independent Variables/Mediator Variables

Task. Subjects were randomly assigned to one of two tasks: in the unaided task, subjects could not write down information as they searched, while those in the aided task received a blank sheet of paper divided into unlabeled rows and columns. They also received written instructions to enter all the brand/attribute information acquired during the search. To ensure that subjects understood the instructions, the researcher showed them how to label and enter their first piece of information. The computer screens were identical in both search conditions, except that those in the aided condition reminded the subject to make the described entries. One young subject who failed to enter the acquired information was eliminated from further analysis.

Age. We recruited subjects from church groups, university staff, and an activity center for senior citizens. Participants ranged from 22 to 88 years old. We divided subjects into two groups based on a median split of the

age distribution (at age 62). The average age for the younger adults was 36 years; the average for elderly adults was 75.

Working-Memory Capacity. We used an established procedure to measure verbal working-memory capacity (see Daneman and Carpenter 1980). We compiled 60 unrelated sentences, 13–16 words in length, using common words. An example sentence is: When I finally reached the seaside, I took a breath of the fresh air. Each sentence was typed on a 3×5 index card in large type. These cards were divided into three sets of 20 cards each. Within each set, the cards were arranged in trials. The first trial had two sentences, the second trial had three sentences, the third trial had four sentences, and so on. Each pair of consecutive trials was separated by an index card containing the word “RECALL.” The cards were presented (to each subject) in a sequence, one card at a time; each time a card was shown, the subject read aloud the sentence contained on it. As soon as the participant had finished reading the sentence on the first card, the researcher placed a second card with a new sentence on top of the first.

The procedure continued until a card with the word “RECALL” on it appeared. At that point the participant had to recall the last word of each of the preceding sentences in that trial. (In the example above, the subject would have to remember the word “air”). The number of sentences in a given trial increased as long as the previous trial was correct. The number recorded for each set was the maximum number of words correctly recalled. A set terminated as soon as the subject made a mistake in recalling a word. We used as a measure of verbal working memory the maximum number of words recalled out of the three sets. This number ranged from one to six, with an average of 3.5 overall. Although this measure is reliable and has been frequently used, it taps only the verbal (i.e., ignores the symbolic or logical) dimension of working memory.

Dependent Variables

The operational definitions of each variable for search intensity and outcome are displayed in Table 1.

Results

Preliminary Data Analyses. We compared elderly and younger adults on several demographic and attitudinal variables. Based on a single-item measure, younger adults had completed more schooling than elderly adults ($F(1,66) = 9.77, p < .05$). Although elderly adults scored higher on our measure of general knowledge about cereal (five items; coefficient alpha = .79) than younger adults ($F(1,74) = 12.34, p < .05$), there were no significant age differences on a test designed to assess awareness and comprehension of nutritional information normally displayed on a cereal box.

While all subjects were able to apply the decision rule three times in a row correctly, elderly adults needed more trials to learn the decision rule than younger adults (young adults, 3.46; elderly, 4.37; $F(1,79) = 8.75, p < .05$).

There were no significant age differences on motivation to process nutritional information (two items; interitem correlation: .78). We ran an ANOVA using search time as a dependent variable. Only a significant task effect emerged (age effect: $F(1,79) = 2.03, p < .16$; task effect: $F(1,79) = 24.64, p < .001$; age-by-task interaction: $F(1,79) = 1.54, p < .22$). Subjects spent more time searching in the aided condition than in the unaided condition, presumably because writing the information down took longer.

Satisficing Behavior. In this study, we defined a person as a satisficer if s/he selected the first brand that met the decision rule. Although this definition varies from the first study, it is more appropriate because in the laboratory experiment all participants were given the decision rule.⁶ Table 2 contains the proportion of satisficers in each condition of our experiment. Chi-square analysis revealed no significant task effect ($\chi^2 < 1$); there was, however, a significant age effect ($\chi^2 = 3.87, df = 1, p < .05$).

Search Intensity. The cell means for this analysis are reported in Table 2, the ANOVA results in Table 3.

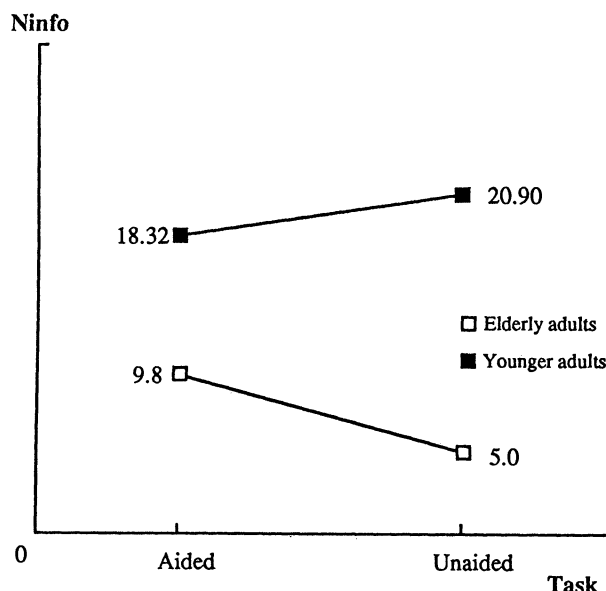
The multivariate analysis of variance (MANOVA) for the number of pieces of information requested, the number of attributes considered, and the number of brands inspected reveals a significant age effect (Wilks's lambda = .60, $F(3,73) = 16.29, p < .001$), a significant task effect (Wilks's lambda = .86, $F(3,73) = 3.9, p < .012$) and a significant age-by-task interaction (Wilks's lambda = .91, $F(3,73) = 2.43, p < .07$). We then inspected the univariate ANOVAs and found a significant age-by-task interaction for each dependent variable.

These interactions, diagrammed in Figures 5–7, show that, for younger adults, the task manipulation did not significantly affect the total pieces of information acquired, the number of attributes they considered, or the number of brands they inspected, while elderly adults in the aided condition requested more pieces of information, searched on more attributes, and inspected more brands than elderly adults in the unaided condition.

We also computed the age effect size in each condition, as indexed by omega squared values (see Winer 1971, pp. 428–430). In the *unaided* condition, age accounts for 37 percent of the variance in the number of pieces of information acquired, 31 percent of the variance in the number of different attributes considered,

⁶Our two studies use slightly different definitions for a satisficer and for search-intensity variables. In each study, we selected definitions that permitted us to compare across conditions within the study, even if it meant sacrificing some cross-study comparability.

FIGURE 5

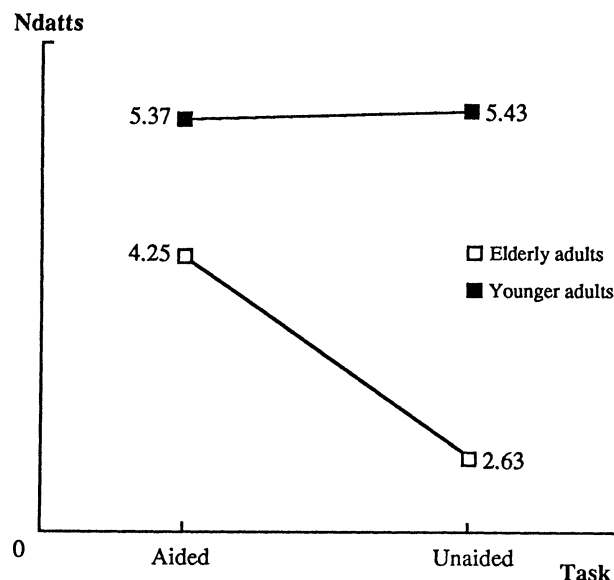
STUDY 2: AMOUNT OF INFORMATION ACQUIRED
AS A FUNCTION OF AGE AND TASK

and 42 percent of the variance in the number of brands inspected. In the *aided* condition, however, these figures decrease to 19 percent, 6 percent, and 7 percent, respectively. Thus, the effect of age on search intensity measures depends on the task conditions.

We considered whether ceiling effects could explain why the aid did not increase search intensity for the younger adults. This possibility was ruled out for the number of different attributes variable because only 5 percent of the younger adults inspected all nine attributes. Because the results were not so clear-cut for the other search-intensity variable with an upper bound (i.e., the number of brands inspected), we ran another ANOVA analysis using the number of search-switches to a different brand as a dependent variable with no upper bound. The results were remarkably similar to the previous analysis (age effect: $F(1,79) = 15.6$, $p < .05$; age-by-task interaction: $F(1,79) = 5.2$, $p < .02$, with brand search-switches increasing in the aided condition for elderly adults but not for younger adults).

Age, Working-Memory Capacity, and Search Intensity. As predicted in Hypothesis 4, working-memory capacity correlated negatively with age ($r = -.42$, $p < .007$). We tested through correlation analysis whether working-memory capacity mediated age effects. In the unaided condition only, where working memory was expected to have the maximum impact, we correlated the respondent's age with the three search-intensity measures. The correlation coefficients between age and search-intensity measures are $-.69$ for the number of pieces of information collected, $-.40$ for the number of attributes considered, and $-.64$ for the number of

FIGURE 6

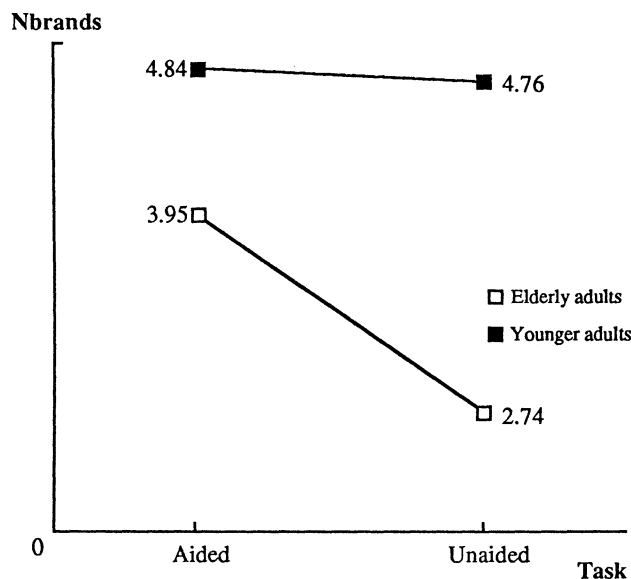
STUDY 2: NUMBER OF DIFFERENT ATTRIBUTES INSPECTED
AS A FUNCTION OF AGE AND TASK

brands inspected. There was almost no attenuation in these three correlations when the influence of working-memory capacity was partialled out (the three partial correlations, in respective order, were $-.64$, $-.27$, and $-.57$).

Thus, our measure of working-memory capacity does not significantly mediate age effects on search intensity. Nonetheless, we pursued the question of how working-memory capacity, independent of age, influences search intensity. Subjects were divided into three equal groups, based on working-memory capacity scores: low ($wm \leq 2$), medium ($2 < wm \leq 3$), and high ($wm > 3$). We then compared the scores of these groups on the search-intensity measures in the aided and unaided conditions. The low group searched significantly more attributes in the aided condition (5.46) than in the unaided condition (3.25; $p < .01$). But the aid did not affect the medium group (4.38 vs. 4.58) or the high group (4.67 vs. 4.43). There were no significant differences between groups in the aided condition, but in the unaided condition there were significant differences between low and medium groups ($p < .06$) and between low and high ($p < .10$) groups. The same pattern of results held for the number of brands inspected, and for the number of pieces of information acquired. Taken together, these results reveal that (1) when working-memory capacity is low, it adversely affects search intensity and (2) instructions to write information down help those with low working-memory capacity.

Search Outcome. For satisfaction, the ANOVA results (Table 3) indicate a significant age effect: younger adults are more satisfied with their choice than elderly adults (22.39 vs. 19.98). Chi-square tests indicate that

FIGURE 7

STUDY 2: NUMBER OF DIFFERENT BRANDS INSPECTED
AS A FUNCTION OF AGE AND TASK

elderly adults made fewer correct choices than younger adults ($\chi^2 = 3.24$, $df = 1$, $p < .07$) in both the aided and unaided conditions.

The Relationship between Search Intensity and Search Outcome. As anticipated in Hypothesis 2, consumer satisfaction increased (1) as the total pieces of information collected increased ($r = .19$, $p < .10$), (2) as the number of attributes considered increased ($r = .19$, $p < .10$), and (3) as the number of brands inspected increased ($r = .26$, $p < .02$). Apparently when consumers consider unfamiliar brands, their satisfaction increases as their search becomes more intense.

Summary

In this study we found that elderly adults tend to satisfice more and search less intensely and less accurately than younger adults, especially in an environment in which information has to be stored and processed in memory. Because familiar brand names were not used, we can rule out prior brand knowledge as an explanation for these findings. When we asked subjects to write the information down, age effects were greatly diminished for the search-intensity measures but not for the search-outcome measures. These results suggest that, when consumers are solving new search problems, differences in information-processing ability may explain age differences in search.

DISCUSSION

The results from the field and laboratory studies yield some insights about nutritional information usage and

also about age differences in consumers' search for and use of the information. In the field study, 29 percent of the shoppers in the unconstrained condition gave nutrition as a reason for their cereal choice. Yet shoppers looked at only 5 percent of the nutritional panels on the cereals they inspected. There are several possible explanations for this disparity. For instance, consumers may believe that all cereals have equal and acceptable nutritional levels; they may also know nutritional quality from previous experience; or they may use information not contained in the nutritional label (such as appearance and taste) to form judgments.

Although in the course of normal shopping there are few age differences in the way people shop for cereal, some differences emerge in the constrained field condition and in the laboratory. We found that individually delivered reminders to use nutritional information can promote more extensive searching and increase the use of nutritional panels, especially with younger adults and especially when delivered at the point of purchase. Elderly adults, however, even when given specific instructions, searched less intensely and made less accurate choices than younger adults. Even though they appeared to understand the decision rule, the elderly consumers had difficulty applying it in a noisy environment in which information on many attributes (in addition to the critical sodium and fiber attributes) was available. In contrast, elderly subjects assigned to the aided condition of our laboratory study engaged in more extensive deliberation, evaluated important attributes, and expended more effort (when compared with elderly subjects in the unaided condition).

Choice Accuracy

One could argue that the elderly consumers were consistently less accurate on choices than younger adults because they may have failed to understand the decision rule. However, in both our studies, consumers were able to apply the rule before beginning the shopping task. In addition, data from a debriefing interview in the supermarket study indicate that elderly subjects were as likely as younger subjects to *think* that they had correctly applied the decision rule.⁷ Thus, it appears inappropriate to ascribe the choice accuracy results of elderly subjects to their lack of understanding of the decision rule. Rather, elderly adults apparently encountered problems obtaining and integrating data about both attributes when this critical information was embedded in other product information (in the field study) or when information on different attributes was presented at different times (in the laboratory study).

⁷As part of a debriefing exercise, subjects assigned to the constrained task condition were asked to recall the level of sodium and fiber in their chosen cereal. About 90 percent of both age groups estimated that there were less than 200 milligrams of sodium and at least 2 grams of fiber in a serving.

Such difficulties may arise from an age-related decrease in ability to discriminate between relevant and irrelevant information, perhaps because of increasing field dependence (see Witkin and Goodenough 1981), and/or an age-associated decrease in ability to integrate information, perhaps because of a general decline in cognitive functioning (Kausler 1982). One of our findings suggests that difficulty in integrating information may be the primary problem: if we isolate just the *first* criterion (sodium content), most of the older and younger subjects in both studies accurately selected a cereal that had less than 200 milligrams of sodium. The failure of our aid to reduce age differences in choice accuracy also points to age-related changes in the ability to integrate information, for the aid simply provided an external record of information without directing processing or promoting information integration.

Search Intensity

One could also argue that the age differences in search intensity are simply a result of the elderly shoppers' failure to apply the decision rule correctly. This explanation lacks empirical support, however; when we repeated our analyses after excluding all subjects who did not select a correct cereal, the pattern of results for search intensity did not change in either study. Instead, we think that elderly consumers searched less intensely than younger adults in part because of a limited working-memory capacity. We found a relatively high incidence of low scores on our working-memory measure among elderly subjects. As our analysis of working-memory data suggests, when this memory falls below a minimum level, it may impair one's ability to conduct an extensive search.

Even in the aided task, age accounted for some of the variance in search intensity, and age differences in satisficing persisted. Like other decision makers who face processing constraints, elderly adults may use a satisficing strategy to reduce the cognitive effort in solving problems (Janis and Mann 1977).

Limitations

Some limitations of our research restrict our ability to generalize the findings. First, in both studies we devised elaborate screening procedures because we wanted to rule out potential problem areas (e.g., eyesight deficiency, failure to understand the decision rule, and inability to learn how to operate the computer) as explanations for our findings. Therefore, our results may not generalize to groups who are not as cognitively and physically able as those who withstood our screening procedures. Second, like most developmental researchers, we attribute differences between older and younger consumers to age-related changes in information-processing abilities and not to birth cohort. Although cohort differences such as subjects' educational experi-

ences may explain some of our results, these confounds were not considered because objective measures are lacking. Third, we relied mostly on observational and computer recording methods; while these methods provide detailed and reliable information about actual behavior, their potential to aid our understanding of cognitive processes is indeed limited.

Finally, our measure of working-memory capacity needs to be refined in future work. For instance, there is some evidence that there is no single entity called working memory but, rather, a number of memories that are domain specific (Light 1991). If this is so, the relatively small attenuations in the correlations between age and the search-intensity variables (when the effects of working memory were statistically controlled) could be due to a mismatch between the domain tapped by our measure of working memory and the domain relevant for our search task.

Contributions

Despite the limitations, this research makes several useful contributions. An important strength of our study is its attention to both external validity (in the field setting) and internal validity (in the laboratory setting). From a methodological perspective, the convergence of findings from the constrained condition in the field and the unaided condition in the laboratory is noteworthy. This convergence indicates that the age differences that we observed are not artifacts of any particular method but reflect real differences in the ability to acquire and use nutritional information in new ways. Clearly, future research on nutritional information usage should continue to use multiple methods to address unanswered questions; developmental research too could benefit from additional studies that investigate parallel problems in different settings.

From a theoretical perspective, our work extends prior information-processing models of nutritional information usage to a broader age range of consumers (Moorman 1990). While Moorman (1990) finds that age is positively related to the ability to process and use nutritional information, our results suggest the opposite. However, the two studies differ greatly in the age of the subjects used: Moorman (1990) uses employed consumers (presumably under the age of 65), while we use consumers ranging in age from their twenties to their eighties.

Our research also sheds light on how consumer age and task familiarity affect satisficing behavior. Older consumers, compared to younger consumers, appear especially likely to adopt satisficing strategies. In addition, both younger and older adults appear to adopt this strategy when making routine decisions in the supermarket.

At a substantive level, this research serves to integrate two research streams allied with the search literature. First, the *economic* orientation (see Ratchford 1982)

emphasizes a cost-benefit trade-off whereby the optimal amount of information is searched when the marginal cost of searching equals the marginal benefits that one accrues from searching. Second, the *information overload* paradigm (Malhotra 1982) emphasizes certain human processing constraints that limit the value of information search beyond a certain point. More specifically, it asserts that there are finite limits on the amount of information that consumers can process effectively; exceeding these limits can yield dysfunctional consequences and incorrect purchasing decisions. Our research suggests that age-related processing abilities or dispositions (such as information-integration skills, working-memory capacity, the tendency to satisfice) may impose natural constraints on how effectively a consumer searches for information.

Public Policy Implications

From a public policy perspective, these studies raise an important question: as new nutritional information becomes available on grocery items, will consumers of any age spontaneously use it to make better decisions? The answer appears to be no, unless the increase in the amount of label information is accompanied by effective consumer education efforts. Our studies show that younger consumers can accurately modify their shopping behavior when specifically asked to use the available information. Thus, there is a need to educate these consumers about how to make wise nutritional choices. Although some of this education may come from the food industry itself, health-care workers can tailor effective nutritional messages to the needs of the consumer (e.g., a hypertensive will receive a different message than a diabetic). In-store, point-of-purchase messages may simply need to remind the shopper to use the available label information.

Our studies also show that, in contrast to younger shoppers, older consumers find it difficult to use label information even when given specific instructions to do so. Because writing information down improved their use of nutritional information, consumer education efforts aimed at elderly adults should not only inform but should also emphasize the benefits of using simple paper-and-pencil decision aids, such as writing down key pieces of information.

In our research, elderly consumers encountered difficulties when they were required to integrate several pieces of information (presented as verbal text) in order to make a brand choice decision. Policymakers should consider approaches to alleviate this problem. For instance, visual symbols may be used on product packages (in addition to verbal text) to convey "preintegrated" nutritional information to consumers. That is, a given visual symbol can designate a unique and nutritionally relevant combination of attributes and levels, and thereby spare elderly consumers the need to integrate verbal information across all these attributes. Moreover,

visual symbols are relatively simple to process and remember, and consumers can learn them easily.

In conclusion, a basic message from our research is that regulatory efforts should not only focus on requiring disclosure of all nutritionally pertinent product information but also should pursue strategies that facilitate utilization of this information by consumers. More specifically, our results suggest that consumer education tailored to the nutritional needs and processing abilities of different age segments can improve information use and decision quality.

[Received September 1990. Revised November 1992.]

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